

HW 6.5.1: Parametric Equations – Projectile Motion

$$x(t) = (v_0 \cos \theta)t$$

$$y(t) = h_0 + (v_0 \sin \theta)t - 16t^2$$

1. Partnering up with Tiger Weeds is Jordan Spittoon out of the Dallas area. He hits a shot that goes 280 feet and skims the top of a 120-foot tree at the peak of the ball's path.
 - a. Use regression and/or matrices to create an equation for the path of the ball, $h(x)$.
 - b. Write the equation for the golf ball's path using the vertex form of the equation for a parabola.
 - c. What is the angle at which the golf ball takes off?
 - d. In order to introduce the parameter of time, t , into this situation, find the time it takes the ball to fall to the ground from its maximum height. ($h(t) = -16t^2$, gravity using feet)
 - e. Find the ball's speed when it hits the ground.
 - f. Write a parametric relation for the ball's path that incorporates time and relates this parameter to the distance and height that the ball travels.

2. A baseball is hit when it is 3 feet above the ground and leaves the bat with initial velocity of 150 feet per second and at an angle of elevation of 20° .
 - a. Write a parametric equation that relates time to the horizontal distance and height of the ball.
 - b. There is a fence that is 400 feet from home plate. How long does it take the ball to reach the fence?
 - c. The fence is 20 feet tall; does the ball clear the fence earning the batter a homerun?
3. A golfer hits a ball with an initial velocity of 133 feet per second and an angle of 36° for the horizontal.
 - a. How long is the ball in the air (hang time)?
 - b. How far does the ball travel horizontally?
 - c. What is the maximum height of the ball?
 - d. If there were a 10 ft/sec headwind, would it affect the horizontal or vertical distance? How? Write a new parametric equation that incorporates the headwind in the equation.

Answers:

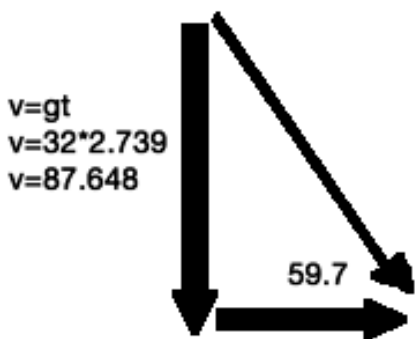
1.

a. $h(x) = -0.006122x^2 + 1.7143x$

b. $h(x) = -0.00612(x - 140)^2 + 120$

c. 59.7°

d. 2.739 seconds



e.

$$\sin(59.7^\circ) = \frac{v_y = 87.648}{v_{\text{landing}}}, \quad v_{\text{landing}} = 101.5 \text{ ft/sec}$$

f. $x(t) = (101.5 \cos(59.7))t$

$$y(t) = (101.5 \sin(59.7))t - 16t^2$$

2.

$$x(t) = (150 \cos(20))t$$

a. $y(t) = 3 + (150 \sin(20))t - 16t^2$

b. $400 = (150 \cos(20))t, \quad t = 2.838 \text{ seconds}$

c. $y(2.838) = 3 + (150 \sin(20))t - 16t^2 = 19.73 \text{ ft. No, it is not a homerun.}$

3.

a. 4.886 seconds

b. 525.73 ft

c. 95.49 ft

d. It will only affect the horizontal component. Since wind is a velocity, it needs to be multiplied by t , and the negative is due to the headwind (a tailwind would have been positive).

$$x(t) = 133 \cos(36)t - 10t$$